AI-Powered Research Gap Identifier & Trend Analyzer

Project ID: R26-SE-023

Project Proposal Report

Wijesinghe L.P.P.

IT22555380

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

AI-Powered Research Gap Identifier & Trend Analyzer

Project ID: R26-SE-023

Project Proposal Report

Wijesinghe L.P.P.

IT22555380

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Wijesinghe L.P.P.	IT22555380	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr. Jeewaka Perera	14-03-2026	
Ms. Poojani Gunathilake	14-03-2026	

ABSTRACT

Managing academic research projects requires effective coordination between students, supervisors, and academic coordinators. In many universities, research activities such as task tracking, document sharing, and communication are still handled using separate tools like email, spreadsheets, and messaging platforms. This fragmented approach makes it difficult to maintain a clear view of project progress and often leads to communication gaps, missed deadlines, and inefficient supervision. As the number of research projects increases in higher education institutions, there is a growing need for structured digital platforms that can support collaboration and progress monitoring throughout the research lifecycle.

Existing systems and project management tools provide certain features such as task management or administrative project tracking. However, most of these systems are either designed for general workplace collaboration or focus mainly on administrative functions rather than active research collaboration. As a result, there is a lack of integrated platforms specifically designed to support both collaboration and structured progress tracking within academic research environments.

To address this issue, this research proposes the development of a Collaborative Workspace and Progress Tracking System designed for academic research projects. The proposed system will provide a centralized digital workspace where students and supervisors can manage tasks, monitor research progress, share documents, and communicate within a single platform. The system will be implemented as a web-based application using React for the frontend, Go with the Gin framework for the backend, and Supabase as the cloud database service.

The expected outcome of this research is a functional prototype that improves transparency, collaboration, and project monitoring in academic research projects. By providing structured progress tracking and collaborative workspace features, the system can support more efficient supervision and better organization of research activities.

From a social perspective, the platform can contribute to improving the quality of research management in universities and support students in completing research projects more effectively. From a commercial perspective, the system has potential to be adopted by educational institutions as a scalable research management platform.

Keywords: Collaborative Workspace, Research Project Management, Progress Tracking, Academic Collaboration, Web-Based System

DEDICATION PAGE

This proposal is dedicated to my parents and family for their constant encouragement and support, and to my academic mentors for their invaluable guidance and commitment to fostering knowledge. I also extend this dedication to the broader research community striving to advance data privacy and security, to whom this work aims to make a meaningful contribution.

ACKNOWLEDGEMENT

I would like to sincerely express my gratitude to supervisors, Mr. Jeewaka Perera and Ms. Poojani Gunathilake, Department of Software Engineering at the Sri Lanka Institute of Information Technology, for their constant support and guidance regarding the development of this research proposal. Their knowledge and constructive feedback have been invaluable in shaping the direction of this project and in addressing possible challenges. I would also like to thank the panel members for their invaluable advice. Indeed, their input was great in pointing out essential improvements to be made and in refining the proposal for effective and feasible study. Finally, I'd like to express my appreciation to everyone who helped me complete this study, whether directly or indirectly. Your invaluable help and suggestions inspired me to achieve this task.

TABLE OF CONTENT

DECLARATION 3

ABSTRACT 4

DEDICATION PAGE 5

ACKNOWLEDGEMENT 6

TABLE OF CONTENT 7

LIST OF TABLES 10

LIST OF FIGURES 11

LIST OF ABBREVIATIONS 12

Abbreviation 12

Meaning 12

LIST OF APPENDICES 13

1. INTRODUCTION 14

1.1 Background and Context 14

1.2 Literature Review 16

1.3 Research Gap 19

1.4 Research Problem 22

2. OBJECTIVE 24

2.1 Main Objective 24

2.2 Specific Objectives 25

3. METHODOLOGY 27

3.1 Research Approach 27

3.2 Technical Methods and Algorithms 27

3.3 Tools and Platforms with Justification 28

3.4 Data Collection and Ethical Considerations 29

3.5 Validation Strategy and Performance Metrics 30

4. HIGH-LEVEL SYSTEM ARCHITECTURE 31

4.1 System Architecture 31

5. USER REQUIREMENTS 37

5.1 Stakeholder Identification 37

5.2 Functional Requirements 37

5.3 Non-Functional Requirements 38

6. COMMERCIALIZATION PLAN 40

6.1 Target Market and Customer Persona 40

6.2 Value Proposition 41

6.3 Revenue Model 41

6.4 Cost Estimation 42

6.5 Pricing Strategy 42

6.6 Competitive Advantage 43

6.7 Intellectual Property Considerations 43

7. BUDGET AND JUSTIFICATION 44

7.1 Hardware Costs 44

7.2 Software Licenses 44

7.3 Overall Budget Consideration 45

8. WORK BREAKDOWN STRUCTURE (WBS) 46

9. GANTT CHART 47

10. REFERENCES LIST 48

11. APPENDICES 50

LIST OF TABLES

Table 1.1- Comparison of Existing Research Systems and Proposed System 21

Table 2 - individual KPI Table 30

Table 3 - Hardware Cost Table 44

Table 4 - Software Licenses Cost 45

LIST OF FIGURES

Figure 1 - Overall System Architecture 31

Figure 2 - Individuwal Component Architecture Diagram 33

Figure 3 - WorkbreakeDown Structure Diagram 46

Figure 4 - Grant Chart 47

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BI	Business Intelligence
CRUD	Create, Read, Update, Delete
DB	Database
FYP	Final Year Project
ICT	Information and Communication Technology
IT	Information Technology
KPI	Key Performance Indicator
LMS	Learning Management System
NLP	Natural Language Processing
PM	Project Management
PMS	Project Management System
PM-PT	Progress Monitoring Prototype Tool
R&D	Research and Development
REST	Representational State Transfeı
SPM	Software Project Management
SQL	Structured Query Language
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
VCS	Version Control System
WIP	Work In Progress

LIST OF APPENDICES

APPENDICES 1- Plagiarism Report 50

1. INTRODUCTION

1.1 Background and Context

In modern universities, research projects are an important part of undergraduate and postgraduate education. These projects often involve collaboration between students, supervisors, and sometimes industry partners. Managing such projects requires regular communication, task coordination, document sharing, and continuous monitoring of progress. However, in many academic environments these activities are still handled using multiple disconnected tools such as email, messaging platforms, spreadsheets, and basic project management applications. This fragmented approach often makes it difficult to maintain a clear view of project progress and collaboration activities.

This problem affects several groups within the academic environment. Students face difficulties when coordinating tasks, tracking deadlines, and maintaining organized documentation throughout the research process. Supervisors also struggle to monitor the progress of multiple student projects simultaneously, especially when updates are shared through informal communication channels. As a result, delays, miscommunication, and lack of transparency can occur during the research process. Since final year research projects are often time-constrained and require structured supervision, the lack of an integrated workspace can negatively affect both project outcomes and learning experiences.

Globally, research collaboration has become increasingly important in higher education. Studies show that collaborative research environments improve knowledge sharing, productivity, and project success. In addition, digital tools have become essential in supporting collaborative work and project monitoring. Despite the availability of several project management tools such as Jira, Trello, and GitLab, many of these platforms are primarily designed for software development teams rather than academic research projects. Existing systems often focus mainly on task management and do not fully support the unique needs of research collaboration, such as structured supervision, research documentation, and continuous progress monitoring. As a result, students and supervisors often rely on a combination of separate tools instead of a single unified system.

Some academic institutions have attempted to introduce project management or academic project information systems to improve research management. While these systems provide basic functions such as project submission and milestone tracking, they usually lack interactive collaboration features, real-time progress visualization, and integrated communication capabilities. In addition, many existing solutions do not provide a dedicated collaborative workspace where research teams can organize their work, share resources, and track progress in a transparent and structured way. These limitations highlight the need for a more comprehensive system specifically designed for academic research collaboration.

To address this problem, this research proposes the development of a Collaborative Workspace and Progress Tracking system designed for academic research projects. The proposed solution aims to provide a centralized digital workspace where students and supervisors can collaborate effectively. The system will support features such as task management, progress tracking dashboards, document sharing, and structured communication within research teams. By integrating these functions into a single platform, the system can help improve transparency, coordination, and efficiency throughout the research process. The platform will be implemented as a web-based application using React for the frontend, Go with the Gin framework for the backend, and Supabase as the database and backend service, enabling scalable and real-time collaboration features.

This research also contributes to broader global goals related to education and digital innovation. In particular, the proposed system aligns with Sustainable Development Goal 4 (Quality Education), which emphasizes improving the quality of higher education and learning environments through the use of technology. By supporting better collaboration and supervision in research projects, the system can help improve students' research skills and academic outcomes. Additionally, the study aligns with research areas related to collaborative systems, digital workspaces, and intelligent project management within the computing and information systems research cluster.

Overall, improving collaboration and progress monitoring in academic research projects is important for both students and supervisors. Developing a dedicated collaborative

workspace platform can provide a more structured and efficient environment for managing research activities, ultimately enhancing the quality and effectiveness of academic research projects.

Research Question:
"How can a collaborative workspace and progress tracking system be designed to improve coordination, transparency, and monitoring of academic research projects between students and supervisors within a centralized digital platform?"

1.2 Literature Review

Digital collaboration platforms and project monitoring systems have become an important area of research in both academic and software development environments. Many studies have attempted to improve the way teams collaborate, track project progress, and manage research activities through digital systems. These studies highlight the importance of structured project monitoring, collaborative workspaces, and visualization tools to support teamwork and improve productivity. However, existing research also shows that most current solutions still have limitations when applied to academic research environments.

Several studies have explored academic project management systems that support the administration of student research projects. For example, research on Academic Project Information Management Systems shows that digital platforms can help automate tasks such as project allocation, supervisor assignment, and evaluation processes [1]. These systems help reduce manual work and improve record management within universities. However, most of these platforms focus mainly on administrative functions rather than supporting real collaboration between students and supervisors. As a result, they provide limited support for shared workspaces, communication, and continuous monitoring of project activities.

Other researchers have proposed systems that enhance academic project monitoring by introducing structured progress tracking mechanisms. One such example is the Project Assessment and Management System (PAMS), which integrates project monitoring with

automated features such as plagiarism detection and similarity checking using natural language processing techniques [2]. The system allows supervisors to monitor research progress through logs and project milestones. Although the system improves transparency in project evaluation, it still lacks interactive collaboration features such as real time workspace interaction and shared document management.

Research has also focused on monitoring tools designed for project based learning environments. Monitoring software developed for project based learning tasks allows instructors to track student contributions and verify project artifacts automatically [3]. This approach reduces the workload of supervisors and improves accountability among students. However, such systems usually rely on predefined project workflows and do not support flexible collaboration environments where students and supervisors can interact continuously.

Another research direction investigates automated progress tracking using development data. The GitMonitor system analyzes activity in Git repositories to monitor project schedules and visualize development progress through Gantt charts and other visual indicators [4]. This method provides objective monitoring based on actual development activity rather than manual reporting. While this approach works well for software development projects, it is limited to environments where project work is managed through Git repositories and therefore may not apply to many academic research projects that involve activities beyond coding.

Visualization based monitoring approaches have also been studied in collaborative software development. DashVis is a system that uses interactive dashboards to visualize project progress using charts such as burndown charts and cumulative flow diagrams [5]. These visualizations help teams understand project status more clearly during meetings and collaborative discussions. Although the system demonstrates the usefulness of progress visualization, the study mainly focuses on professional software development teams and large display environments rather than typical academic research settings.

Researchers have also explored hybrid project monitoring frameworks that combine multiple development tools to create a unified monitoring dashboard. These systems collect project information from platforms such as GitHub, Jira, and other management

tools and display a combined view of project progress [6]. This approach helps reduce manual reporting and provides a clearer understanding of project status. However, the need to integrate several external tools can increase system complexity and may not be suitable for academic environments where users prefer a simple and centralized platform.

In addition to monitoring systems, studies on collaborative workspaces emphasize the importance of shared digital environments that support communication, task coordination, and knowledge sharing. Collaborative workspace platforms allow team members to work together within a common environment where tasks, documents, and project updates can be managed collectively [7]. These systems improve team coordination and make it easier to track project activities. However, most existing collaborative workspace platforms are designed for general teamwork and do not specifically address the needs of academic research collaboration.

Recent research has also examined collaborative learning tools and virtual collaboration environments. Platforms such as Corobolab provide shared environments where students can collaborate on learning activities and AI related projects [8]. These systems demonstrate the potential of digital workspaces to improve collaboration and engagement among students. Nevertheless, these tools are mainly designed for collaborative learning rather than structured research project management.

Studies on collaborative research environments further highlight the importance of integrated digital platforms for supporting interdisciplinary research teams [9]. These systems aim to provide shared communication channels, task coordination tools, and knowledge sharing mechanisms for distributed research teams. While such systems improve collaboration across different locations, many of them lack dedicated project progress tracking features that supervisors can use to evaluate student research activities effectively.

Overall, the literature indicates that many digital systems have been developed to support project management, collaboration, and progress monitoring. Existing research demonstrates the value of dashboards, automated monitoring tools, and collaborative workspaces in improving project coordination and transparency. However, most current solutions address only specific aspects of the problem. Administrative systems focus on

project documentation, monitoring tools concentrate on development data, and collaborative platforms emphasize communication and teamwork. As a result, there is still a lack of integrated systems that combine collaboration, structured progress tracking, and academic research management within a single platform. This limitation highlights the need for a dedicated collaborative workspace system that can effectively support research teams and supervisors in managing academic research projects.

1.3 Research Gap

The literature review shows that many systems and research studies have attempted to improve project management, collaboration, and monitoring in academic and software development environments. Several platforms focus on administrative management of student projects, while others concentrate on automated monitoring, visualization dashboards, or collaborative tools. Although these studies contribute useful features such as progress dashboards, artifact tracking, and task monitoring, there is still a significant gap when these systems are applied to academic research project environments.

One major limitation in current systems is the lack of an integrated collaborative workspace specifically designed for academic research teams. Many academic project management platforms mainly support administrative activities such as project registration, supervisor allocation, and final submission processes [1]. While these features help universities manage project records, they do not provide an environment where students and supervisors can collaborate continuously throughout the research lifecycle.

Similarly, systems such as the Project Assessment and Management System (PAMS) improve monitoring by providing structured logs and automated evaluation features [2]. However, these platforms still rely heavily on milestone based reporting rather than supporting real time collaborative interaction between project members. As a result, communication between students and supervisors often continues through external tools such as email or messaging platforms, which leads to fragmented workflows and reduced transparency in project progress.

Some research has proposed automated monitoring systems that track project progress using development data or artifact verification [3][4]. Although these systems provide objective insights into project activities, they are often designed specifically for software development workflows and rely on development tools such as Git repositories. This approach limits their usability in broader academic research contexts where project activities include literature reviews, data analysis, documentation, and meetings that cannot always be tracked through development tools.

Visualization based monitoring systems also provide useful insights into project progress. For example, dashboard systems such as DashVis use visual indicators like burndown charts and cumulative flow diagrams to help teams understand project status [5]. However, these systems mainly focus on professional software development teams and collaborative display environments rather than academic research projects involving students and supervisors.

Another limitation found in the literature is that many collaborative workspace platforms are general purpose teamwork tools that are not tailored for academic research workflows [7]. While these platforms support communication and task coordination, they usually lack structured mechanisms for research progress monitoring, supervisor feedback, and academic evaluation. Similarly, collaborative learning tools such as Corobolab support team interaction within learning environments but do not provide comprehensive research project management capabilities [8].

Furthermore, some monitoring frameworks attempt to integrate multiple project management tools into a single dashboard [6]. Although this approach provides a unified view of project progress, it often requires complex integration between several external platforms, which may increase system complexity and reduce usability for students and supervisors.

Based on these observations, it can be seen that existing solutions often address only one aspect of the problem, such as project monitoring, collaboration, or administrative management. There is still a lack of a centralized platform that combines collaborative

workspace capabilities with structured research progress tracking specifically for academic research projects.

To address this gap, this research proposes the development of a Collaborative Workspace and Progress Tracking System designed for academic research environments. The proposed solution aims to integrate key features such as collaborative task management, structured progress monitoring, document sharing, and communication tools within a single digital workspace. By providing a unified environment for both students and supervisors, the system aims to improve transparency, coordination, and monitoring throughout the research process.

Therefore, the novelty of this research lies in designing an integrated collaborative workspace platform that directly addresses the specific needs of academic research project management, rather than adapting tools originally designed for general project management or software development environments. This approach aims to provide a more effective solution for supporting collaboration and progress monitoring in higher education research projects.

Table 1.1- Comparison of Existing Research Systems and Proposed System

Feature / Capability	Research 01 Academic Project Information Management System	Research 02 PAMS	Research 03 GitMonitor	Research 04 DashVis	Proposed System
Project registration and academic administration	Yes	Yes	No	No	Yes
Progress monitoring of research projects	Partial	Yes	Yes	Yes	Yes
Real-time collaborative workspace for students and supervisors	No	No	No	Partial	Yes

Task management within research projects	Partial	Partial	No	No	Yes
Integrated document sharing and management	No	No	No	No	Yes
Communication between research team members	No	Partial	No	No	Yes
Visualization dashboards for project progress	No	Partial	No	Yes	Yes
Monitoring of research activities beyond coding tasks	Yes	Yes	No	Partial	Yes
Centralized platform combining collaboration and monitoring	No	No	No	No	Yes
Designed specifically for academic research collaboration	Partial	Partial	No	No	Yes

1.4 Research Problem

Managing academic research projects in universities often involves multiple activities such as task coordination, communication between students and supervisors, document sharing, and monitoring project progress. In many cases, these activities are handled using separate tools such as email, messaging applications, spreadsheets, and general project management platforms. Because these tools are not integrated into a single environment, it becomes difficult for research teams to maintain a clear and consistent view of project progress. This situation often leads to fragmented communication, missed updates, and difficulty in tracking the actual status of research activities.

Supervisors are usually responsible for guiding several student projects at the same time. Without a centralized system, monitoring each project becomes challenging because progress updates are often shared informally through meetings, messages, or documents. As a result, supervisors may not always have a real-time understanding of project development, which can lead to delays in identifying issues or providing timely feedback. Students also face difficulties when organizing tasks, managing research materials, and communicating progress in a structured manner.

Although several project management and collaboration tools exist, most of them are designed for software development teams or general workplace collaboration. These platforms usually focus on task tracking or communication features but do not address the specific requirements of academic research projects, such as structured supervision, research documentation management, and continuous progress monitoring. Similarly, many academic project management systems mainly support administrative functions rather than active collaboration and workspace management.

Therefore, the core research problem is the lack of an integrated collaborative workspace that allows students and supervisors to effectively collaborate, organize research activities, and monitor project progress within a single platform specifically designed for academic research environments. Addressing this problem requires the development of a system that combines collaboration tools, task management, progress tracking, and shared research resources in a centralized digital workspace.

This research aims to address this problem by designing and developing a Collaborative Workspace and Progress Tracking system that provides a structured environment for managing academic research projects. The system is intended to improve transparency, coordination, and efficiency in research collaboration between students and supervisors.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a collaborative workspace and progress tracking system that enables students and supervisors to effectively manage, monitor, and communicate research project activities in real time through a centralized digital platform.

Final year research projects usually involve multiple stakeholders such as students, supervisors, and coordinators, and require continuous monitoring of tasks, milestones, and deliverables. However, many existing academic project management practices still depend on fragmented tools such as spreadsheets, emails, or manual documentation, which makes it difficult to clearly monitor project progress and maintain effective communication among stakeholders.

In addition, several existing software project management tools provide general collaboration and task management features, but they often lack sufficient mechanisms for visualizing project progress, generating meaningful progress reports, or providing clear insights into project status. These limitations reduce their effectiveness in supporting research project monitoring and decision making.

Therefore, this research aims to develop a collaborative workspace that integrates structured progress tracking, communication support, and project monitoring features within a single system. The proposed solution will allow users to record project activities, track milestones, visualize project progress, and share updates with supervisors in a more organized manner.

By implementing this system, the study intends to improve transparency, coordination, and efficiency in managing research projects, while providing a structured environment that supports better monitoring and decision-making throughout the project lifecycle. This objective directly addresses the identified research gap by delivering a practical solution

that enhances collaborative research project management within an academic environment.

2.2 Specific Objectives

The primary focus of this component is to develop a collaborative workspace and progress tracking mechanism that supports effective management of research projects between students and supervisors. The system aims to provide a structured digital environment where project activities, tasks, and deliverables can be organized and monitored throughout the research lifecycle. Through this approach, the system will enable students to record their work progress regularly while allowing supervisors to review updates and provide feedback in a timely manner.

Based on this concept, the next objective is to design a structured progress tracking feature that allows users to update milestones, record research activities, and monitor the completion status of tasks. This mechanism will help maintain transparency in the research workflow and make it easier for supervisors to understand the current state of the project without relying on scattered communication channels such as emails or manual logs. By maintaining a clear record of activities and milestones, the system will help ensure that research work progresses according to the planned timeline.

Another important objective is to introduce visualization and monitoring features that present project progress in an easy-to-understand format. These features may include dashboards, progress indicators, or timeline-based views that allow supervisors and coordinators to quickly identify delays, completed milestones, and ongoing tasks. Such visual insights can improve decision making and help guide students when adjustments are needed during the project development process.

In addition, the system will support collaborative communication between students and supervisors within the workspace itself. This will allow users to share documents, provide comments, and exchange feedback directly through the platform rather than depending on external communication tools. By centralizing these interactions, the proposed solution

aims to improve coordination and maintain a complete record of project-related discussions and updates.

Finally, the developed system will be evaluated to determine its effectiveness in supporting research project management. Feedback from potential users, including students and supervisors, will be collected to understand how well the system improves collaboration, transparency, and progress monitoring compared with existing practices. The evaluation results will help determine whether the proposed collaborative workspace provides a practical solution for improving research project management in an academic environment.

3. METHODOLOGY

3.1 Research Approach

This research adopts the Design Science Research (DSR) approach, which is commonly used in information systems research where a technological artifact is created to solve a real-world problem. The goal of this research is to design and develop a collaborative workspace platform that improves research collaboration and progress tracking within academic environments.

The Design Science approach is appropriate for this study because the research involves identifying a problem in current research project management practices and proposing a system-based solution. The research process follows several stages including problem identification, requirement analysis, system design, implementation, and evaluation. Through this process, the proposed system will be developed and tested to determine whether it improves collaboration and monitoring in research projects.

3.2 Technical Methods and Algorithms

The proposed system will be implemented as a web-based collaborative platform that allows research teams to manage their work in a structured environment. The system will use a modular architecture where different components handle task management, collaboration, and progress monitoring.

Several technical methods will be used within the system:

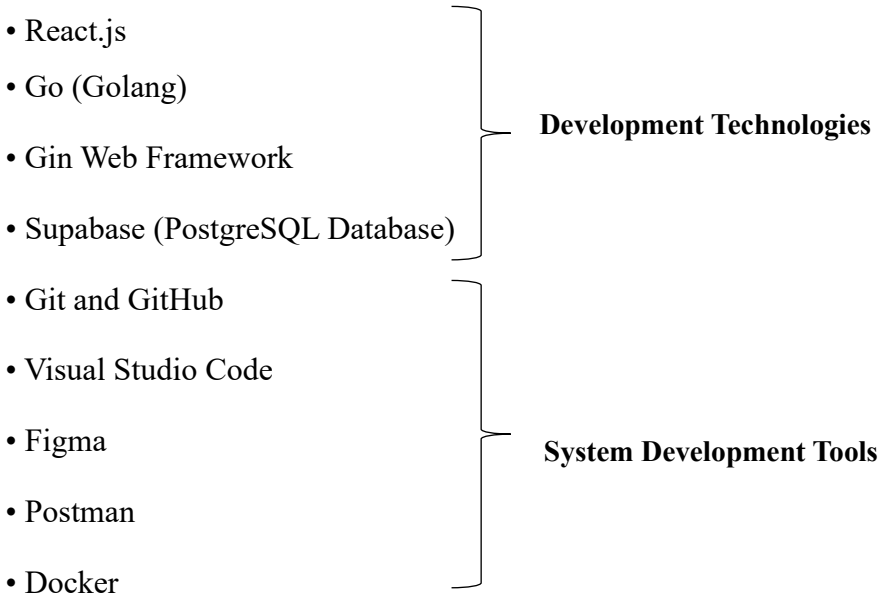
- Task management mechanisms will allow project members to create tasks, assign responsibilities, and monitor completion status.
- Progress tracking algorithms will calculate project progress based on completed tasks, deadlines, and milestone achievements.
- Dashboard visualization methods will display project progress using graphical indicators such as progress bars and activity summaries.

- Role-based access control will be implemented to manage permissions for students, supervisors, and administrators.
- RESTful API communication will enable secure interaction between the frontend and backend services.

These technical methods allow the system to provide structured project monitoring while supporting collaboration between research team members.

3.3 Tools and Platforms with Justification

The proposed Collaborative Workspace and Progress Tracking System will be developed using a set of modern web development tools and technologies that support scalability, reliability, and efficient collaboration. These tools are selected to ensure the system can support real-time project monitoring and collaboration between students and supervisors.
Tools and Technologies



The frontend of the system will be developed using React, a widely used JavaScript library for building interactive user interfaces. React enables efficient component-based development and improves user experience through responsive interfaces.

The backend will be implemented using Go (Golang) with the Gin web framework. Go is known for its high performance, concurrency support, and efficient handling of web services. The Gin framework provides lightweight and fast API development, making it suitable for building scalable backend services.

The system will use Supabase as the backend database platform. Supabase provides a PostgreSQL-based database along with authentication services and real-time data capabilities. This allows the system to store project data securely while supporting real-time updates for collaboration features.

Additional tools such as GitHub will be used for version control and project management during the development process. These tools help ensure organized development and proper tracking of system changes.

3.4 Data Collection and Ethical Considerations

Data required for the research will mainly be collected through literature review and system evaluation activities. The literature review provides information about existing research systems, collaborative tools, and project monitoring platforms. This information helps identify research gaps and define system requirements.

For system evaluation, feedback may be collected from potential users such as students and supervisors who interact with the prototype system. Their feedback will help evaluate usability and system effectiveness.

Ethical considerations will be followed during the research process. Any user data collected during testing will be handled carefully to protect privacy and confidentiality. Participation in evaluation activities will be voluntary, and no sensitive personal data will be stored without proper consent.

3.5 Validation Strategy and Performance Metrics

The developed system will be evaluated using both functional testing and user-based evaluation methods. Functional testing will ensure that all system modules operate correctly. This includes testing features such as task creation, progress tracking, workspace collaboration, and data management.

User-based evaluation will assess how effectively the system supports collaboration and progress monitoring in research projects. Feedback will be collected from users regarding usability, usefulness, and overall system experience.

Several performance metrics will be used during evaluation:

- System response time when performing common tasks
- Accuracy of progress tracking calculations
- Usability and user satisfaction levels
- Reliability of system features during multiple user interactions

These evaluation methods will help determine whether the proposed collaborative workspace system successfully improves research project management and progress monitoring in academic environments.

Self-Evaluation Plan and Key Performance Indicators (KPIs)

To evaluate the effectiveness of the proposed system, several Key Performance Indicators (KPIs) will be used. These KPIs measure both the technical performance of the system and the user experience during collaboration and progress monitoring.

Table 2 - individual KPI Table

Evaluation Aspect	KPI (Metric)	Measurement Method	Target / Benchmark
Task Tracking Accuracy	Accuracy of progress calculations (%)	Compare system progress values with manual calculations	≥ 90%
System Response Time	Average response time (seconds)	Performance testing during task creation and updates	≤ 2 seconds
System Reliability	Successful API request rate (%)	Backend system testing and error log analysis	≥ 95%

Usability	SUS (System Usability Scale) Score	User testing with at least 10 participants	SUS ≥ 80
Collaboration Efficiency	User satisfaction rating (%)	Survey feedback from users	≥ 85% satisfaction

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Architecture

Overall System Architecture

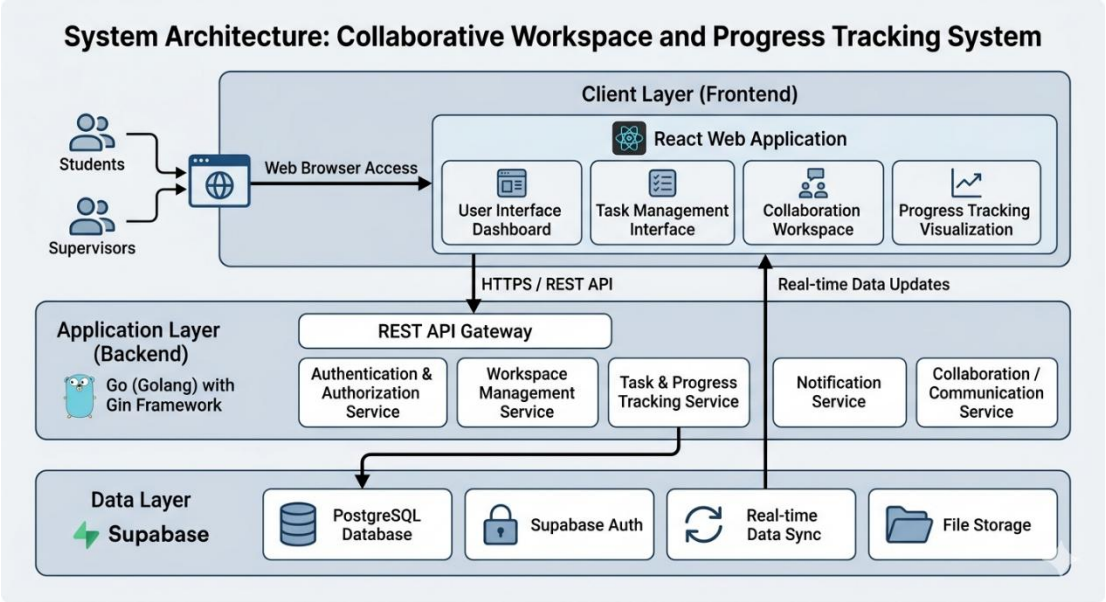


Figure 1 - Overall System Architecture

Figure 4.1 illustrates the overall architecture of the proposed Collaborative Workspace and Progress Tracking System. The architecture is designed as a unified web-based platform that supports collaboration, task management, and progress monitoring for academic research projects. The system mainly connects students, supervisors, and coordinators within a centralized environment so that research activities can be tracked and managed more effectively.

At the first stage, users access the system through a secure web interface. The primary users of the platform are students, supervisors, and project coordinators. After authentication, each user is directed to a personalized dashboard where they can view their assigned research projects, upcoming tasks, deadlines, and recent activity updates. The dashboard serves as the main interaction point of the system and provides a clear overview of project progress.

When a research project is initiated, the system allows students to create project workspaces where research tasks, milestones, and deliverables can be organized. Each project workspace contains a structured task list that represents the different stages of the research process, such as literature review, proposal preparation, implementation, testing, and documentation. Students are able to update the completion status of tasks, upload related documents, and record progress updates within the system. These updates are automatically stored in the database and become visible to supervisors in real time.

Once the project activities are recorded, the Progress Tracking Module processes this information and generates progress summaries for the project. This module analyzes task completion rates, milestone achievements, and activity updates to determine the overall progress of the research work. The system then presents these results through visual dashboards and progress indicators, which help supervisors quickly understand the current status of each project.

The Collaboration and Communication Module enables students and supervisors to exchange feedback within the workspace itself. Supervisors can review submitted documents, provide comments, and guide students through the research process. This feature ensures that all project communication remains organized and recorded within the system rather than being scattered across different communication tools.

Another important component of the architecture is the Notification and Alert System. This module sends reminders and notifications to users when important events occur, such as approaching deadlines, new feedback from supervisors, or updates in project tasks. These notifications help maintain continuous engagement and ensure that students remain aware of their project responsibilities.

All project data, including task records, user information, and progress updates, are stored securely in a centralized database. The database ensures that research information is consistently maintained and can be retrieved whenever required. Access control mechanisms are implemented so that users can only view or modify data relevant to their assigned projects. This approach helps maintain data security and integrity within the system.

Through this architecture, the proposed solution provides a structured and collaborative environment that improves transparency, communication, and monitoring in academic research project management. By combining task tracking, collaboration tools, and progress visualization in a single platform, the system aims to address many of the limitations present in traditional methods of managing research projects.

4.2 Individual Component Architecture

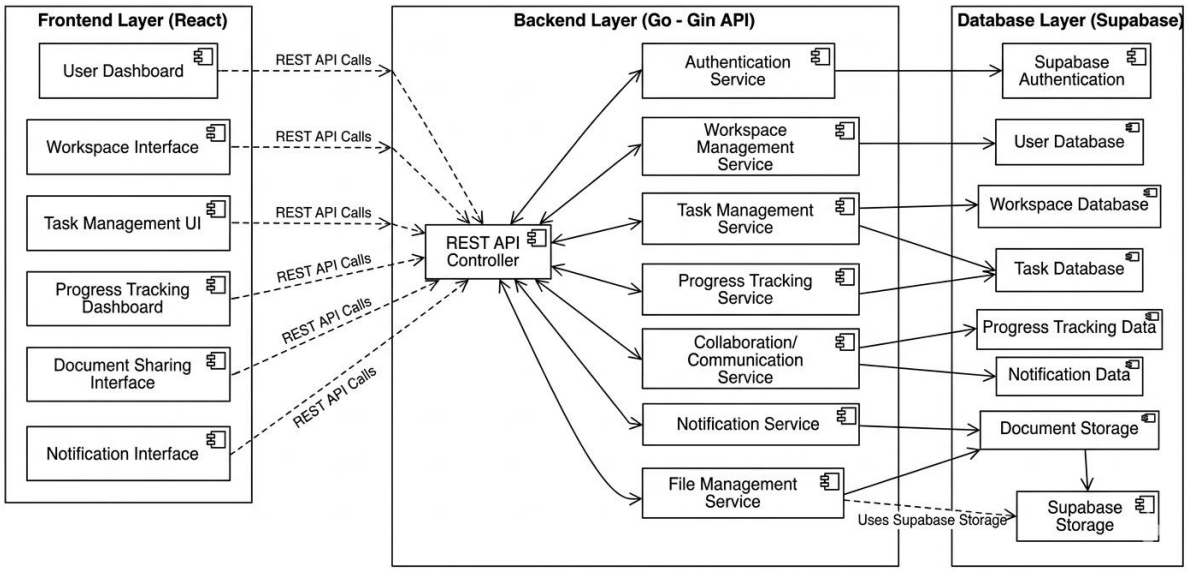


Figure 2 - Individuwal Component Architecture Diagram

The internal working structure of the system can be further explained through the Individual Component Architecture, which divides the system into several specialized functional modules. These modules work together to manage research project activities and ensure smooth interaction between users.

The process begins with the User Management Component, which handles user registration, authentication, and access control. Each user is assigned a specific role, such as student, supervisor, or coordinator. Based on this role, the system grants appropriate permissions that determine which features the user can access within the platform.

After successful login, users interact with the Project Workspace Component. This module allows students to create and manage research projects within the system. Each project workspace includes details such as project title, research area, group members, supervisor information, and project timelines. This structured organization helps maintain clarity and accountability in the project environment.

The Task and Milestone Management Component is responsible for organizing research activities into manageable tasks. Students can create tasks, assign deadlines, and mark tasks as completed once the work is finished. Milestones are used to represent major stages of the research project, and the system monitors whether these milestones are achieved within the expected time frame.

Another important part of the system is the Progress Monitoring Component. This module collects task updates and milestone information to calculate the overall progress of the project. The collected data is analyzed to produce progress indicators, timelines, and summary reports that supervisors can use to evaluate the development of the research work.

The Document and Resource Management Component enables students to upload research documents, presentations, and reports directly into the project workspace. Supervisors can review these documents and provide feedback through the platform. This ensures that all project-related materials remain organized and accessible throughout the research process.

Finally, the Reporting and Visualization Component generates dashboards and visual summaries of project activities. These visualizations present information such as task completion status, milestone progress, and activity logs in a clear and understandable

format. By providing visual insights, the system helps supervisors and coordinators quickly identify potential delays or issues in research projects.

Through the integration of these components, the system ensures that research activities are systematically recorded, monitored, and communicated among stakeholders.

The system components interact through RESTful API communication between the frontend and backend layers. When users perform actions through the React web interface, requests are sent to the backend API developed using the Go Gin framework. The backend services process these requests by interacting with different modules such as workspace management, task tracking, and notification services. After processing the request, the system retrieves or updates information stored in the Supabase database and returns the updated data to the user interface. This architecture ensures smooth interaction between system components while maintaining modular system design.

The data flow within the system begins when users interact with the web application through their browser. User requests such as creating tasks, uploading documents, or updating progress are sent to the backend through REST APIs. The backend services validate the request, process the data, and store it in the centralized database. The progress monitoring module then analyzes task updates and milestone information to calculate project progress. The processed data is finally returned to the frontend interface where it is visualized through dashboards and progress indicators for students, supervisors, and coordinators.

Scalability and Security Considerations

The system architecture is designed to support scalability by using a modular service-based structure. The backend services can be extended independently as the number of users or research projects increases. Cloud-based database services provided by Supabase allow the system to handle large volumes of data while maintaining performance. The use of REST APIs also enables future integration with other academic systems such as learning management systems.

Security is an important consideration in the proposed architecture. User authentication and role-based access control mechanisms ensure that only authorized users can access project information. Students can only view and update their assigned project workspaces, while supervisors and coordinators have broader monitoring privileges. Secure communication between the frontend and backend services is implemented using HTTPS protocols to protect sensitive research data.

5. USER REQUIREMENTS

The goal of the system is to support effective collaboration and structured progress monitoring in academic research projects. The requirements were identified based on the research problem, literature review, and analysis of existing systems.

5.1 Stakeholder Identification

The system will be used by several stakeholders who participate in academic research projects. Each stakeholder interacts with the system in different ways depending on their role and responsibilities.

1. Students

   Students are the main users of the system. They will use the platform to create and manage their research project workspaces, organize tasks, upload research documents, and update project progress. The system will help students keep track of deadlines and communicate with supervisors throughout the research process.

2. Supervisors

   Supervisors are responsible for guiding and monitoring student research projects. They will use the system to review project progress, provide feedback on submitted documents, track milestone completion, and monitor multiple projects at the same time. The platform will allow supervisors to have a clear view of the status of each project they supervise.

3. Project Coordinators

   Project coordinators are responsible for overseeing research project activities within the department or faculty. They will use the system to monitor overall project progress, manage project allocations, and ensure that research activities follow institutional guidelines.

5.2 Functional Requirements

Functional requirements describe the main features that the system must provide in order to support collaboration and project monitoring.

FR1: The system shall allow users to register and log in using secure authentication mechanisms.

FR2: The system shall support role-based access control for different users such as students, supervisors, and coordinators.

FR3: The system shall allow students to create and manage research project workspaces.

FR4: The system shall allow users to create, assign, and manage research tasks and milestones within a project.

FR5: The system shall allow students to update the status of tasks and record progress updates.

FR6: The system shall provide supervisors with the ability to monitor project progress through dashboards and progress indicators.

FR7: The system shall allow users to upload and manage research documents and project files.

FR8: The system shall allow supervisors to review submitted documents and provide feedback to students.

FR9: The system shall support communication between students and supervisors within the project workspace.

FR10: The system shall generate visual dashboards that display project progress and milestone completion.

FR11: The system shall send notifications and alerts when deadlines approach or when project updates occur.

FR12: The system shall allow project coordinators to monitor multiple research projects within the system.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance expectations of the system.

- Security

  The system must ensure that user data and research information are protected. Authentication and role-based access control mechanisms will restrict access to

authorized users only. Secure communication protocols will be used to protect data during transmission.

- Performance

  The system should respond quickly to user actions such as loading dashboards, updating tasks, and retrieving project data. The average system response time should remain within acceptable limits to ensure smooth user interaction.

- Usability

  The platform should provide a user-friendly interface that allows students and supervisors to easily navigate the system and manage research activities without requiring advanced technical knowledge.

- Scalability

  The architecture should support an increasing number of users and research projects without significant performance degradation. The use of cloud-based services and modular system design will help support future system growth.

- Reliability

  The system must ensure consistent availability and reliable operation during project activities. System errors or failures should be minimized to prevent disruption to research work.

6. COMMERCIALIZATION PLAN

The proposed component has potential not only as a research prototype but also as a practical digital platform that can be adopted by universities and educational institutions. This section describes how the system can be introduced into the market, the potential users, and the strategies that can support its sustainability as a product.

6.1 Target Market and Customer Persona

The primary target market for this system is higher education institutions, including universities, technical institutes, and research centers that manage undergraduate and postgraduate research projects. Many universities currently rely on manual processes, spreadsheets, or disconnected tools to manage research projects. Therefore, institutions that want to improve the efficiency of their research supervision processes represent a strong potential market for this system.

Within these institutions, the main customer personas include:

- University Administrators and Faculty Coordinators –
  These users are responsible for managing research projects within departments or faculties. They need systems that help monitor project progress, manage project submissions, and ensure that research activities follow academic guidelines.
- Research Supervisors
  Supervisors manage multiple student projects at the same time. They require tools that allow them to track progress, review documents, and communicate with students in a structured environment.
- Students and Research Teams
  Students working on final year research projects need a platform that helps them organize tasks, track milestones, share documents, and receive feedback from supervisors.

6.2 Value Proposition

The proposed system provides a centralized collaborative environment for managing academic research projects. Unlike many general project management tools, this platform is specifically designed for academic research workflows.

- The key value provided by the system includes:
- A unified workspace for research collaboration between students and supervisors
- Structured tracking of research tasks, milestones, and project progress
- Real-time dashboards that allow supervisors to quickly monitor multiple projects
- Centralized storage for research documents and communication records
- Automated notifications and reminders that help maintain project timelines

By integrating these features into a single platform, the system reduces the need to use multiple disconnected tools and improves transparency and coordination within research teams.

6.3 Revenue Model

The commercialization strategy for the system can follow a Software-as-a-Service (SaaS) model. Under this model, educational institutions subscribe to the platform and access the system through a web-based service.

Possible revenue streams include:

- Institutional subscriptions where universities pay an annual or monthly fee to use the platform
- Premium feature packages for advanced analytics, reporting tools, or integrations with learning management systems
- Custom implementation services for institutions that require system customization or integration with existing academic systems

This model allows institutions to use the system without needing to manage their own infrastructure.

6.4 Cost Estimation

The main costs associated with deploying and maintaining the platform include several technical and operational components.

- Infrastructure costs
  include cloud hosting services, database management, and storage for project documents. These services ensure that the platform remains accessible and scalable as the number of users increases.

- Development and maintenance costs
  involve continuous improvement of the system, fixing software issues, and updating features based on user feedback.
- Operational costs
  may include technical support services, system monitoring, and security management to ensure reliable platform operation.

Using cloud-based services and open-source technologies helps reduce development and infrastructure costs, making the platform more financially sustainable.

6.5 Pricing Strategy

A flexible pricing strategy can help encourage adoption by educational institutions. One possible approach is a tiered pricing model based on the number of users or projects.
For example:

- A basic plan designed for small departments with limited research projects.
- A standard plan suitable for medium-sized faculties with multiple research teams.
- An enterprise plan for large universities that manage a large number of research projects across different faculties.

Providing a limited free trial or academic pilot program can also help universities evaluate the platform before making a full adoption decision.

6.6 Competitive Advantage

The main competitive advantage of the proposed system is that it is specifically designed for academic research project management. Many existing platforms such as general project management tools focus mainly on business or software development environments. These tools often lack features that are important for research supervision, such as structured milestone tracking, supervisor feedback systems, and academic evaluation workflows.

The proposed platform integrates collaboration tools, progress monitoring dashboards, document management, and communication features into a single environment tailored for academic use. This specialization provides a stronger fit for universities compared to general-purpose project management systems.

6.7 Intellectual Property Considerations

The intellectual property of the system includes the software design, system architecture, algorithms used for progress tracking, and the user interface design. These components may be protected through software copyright and licensing agreements.

If the system is further developed into a commercial product, the development team or institution may consider registering the platform as a software product or technology innovation under relevant intellectual property regulations. Proper licensing agreements can also define how institutions are allowed to use the system while protecting the ownership of the developed software.

7. BUDGET AND JUSTIFICATION

The budget mainly covers the technical resources and services required to design, implement, and evaluate the system. Since this project is developed as part of an academic research project, the costs are estimated based on essential resources such as hardware, software tools, cloud infrastructure, and development effort.

7.1 Hardware Costs

The system development requires computing devices capable of running development tools, code editors, and testing environments. Most development tasks such as frontend and backend implementation, database management, and system testing will be performed using personal computers or laptops. A reliable computer system is necessary for software development activities such as programming, debugging, and running development frameworks. Stable internet connectivity is also required to access cloud services, collaborate through version control platforms, and test the web-based application environment.

Table 3 - Hardware Cost Table

Hardware Item	Estimated Cost	Description
Development Laptop / Computer	LKR 250,000	Used for coding, system design, testing, and documentation.
Internet Connectivity	LKR 5,000 (monthly)	Required for accessing cloud services, code repositories, and testing the web application.

7.2 Software Licenses

Mainly I uses open-source technologies, which helps reduce software licensing costs. However, some optional tools and services may require subscriptions for advanced features. Most tools selected for this project are open-source or provide free versions

suitable for academic development. This approach minimizes software costs while still providing powerful development tools.

Table 4 - Software Licenses Cost

Cloud Service	Estimated Cost	Description
Supabase Cloud Database	Free / LKR 5,000 monthly (optional upgrade)	Used for database storage, authentication, and real-time data synchronization.
Cloud Hosting Services	LKR 5,000 monthly	Used to deploy and run the web application for testing and demonstration.
File Storage Services	Included in Supabase	Used to store research documents uploaded by users.

7.3 Overall Budget Consideration

The overall budget for this project remains relatively low because most of the technologies used are open-source and cloud services provide free tiers suitable for research and prototype development. This cost-efficient approach ensures that the system can be developed and tested effectively without requiring significant financial investment.

8. WORK BREAKDOWN STRUCTURE (WBS)

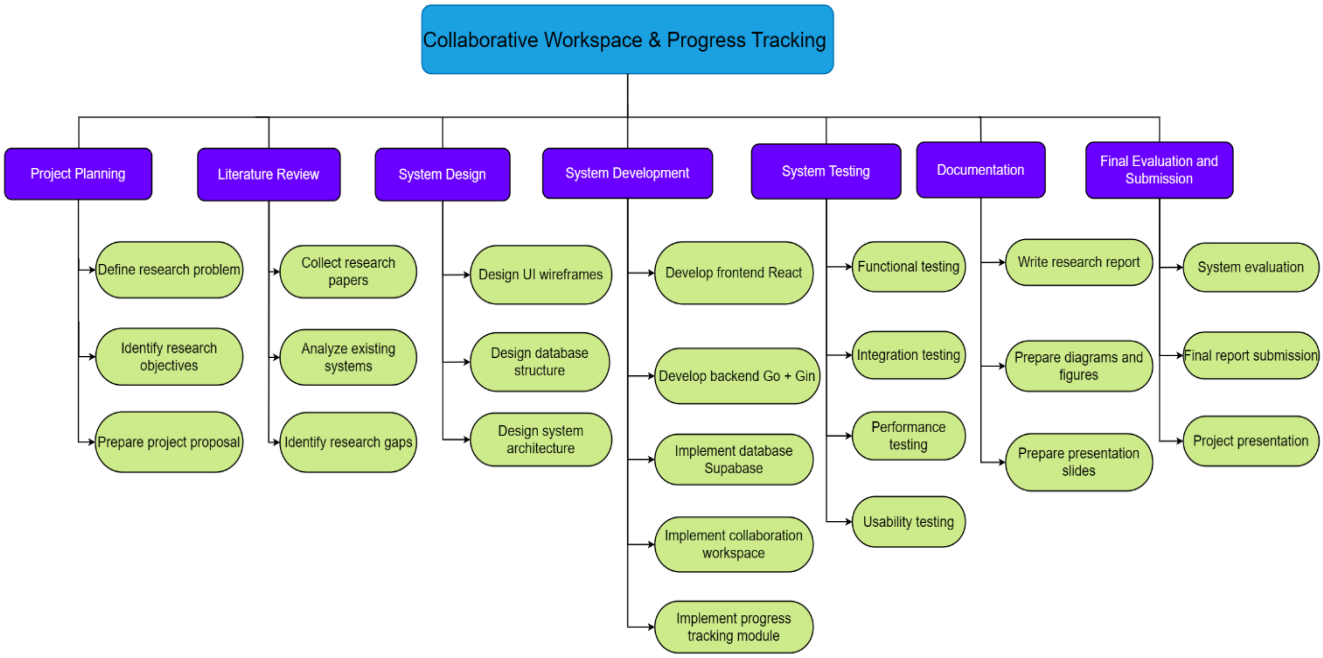


Figure 3 - WorkbreakeDown Structure Diagram

9. GANTT CHART

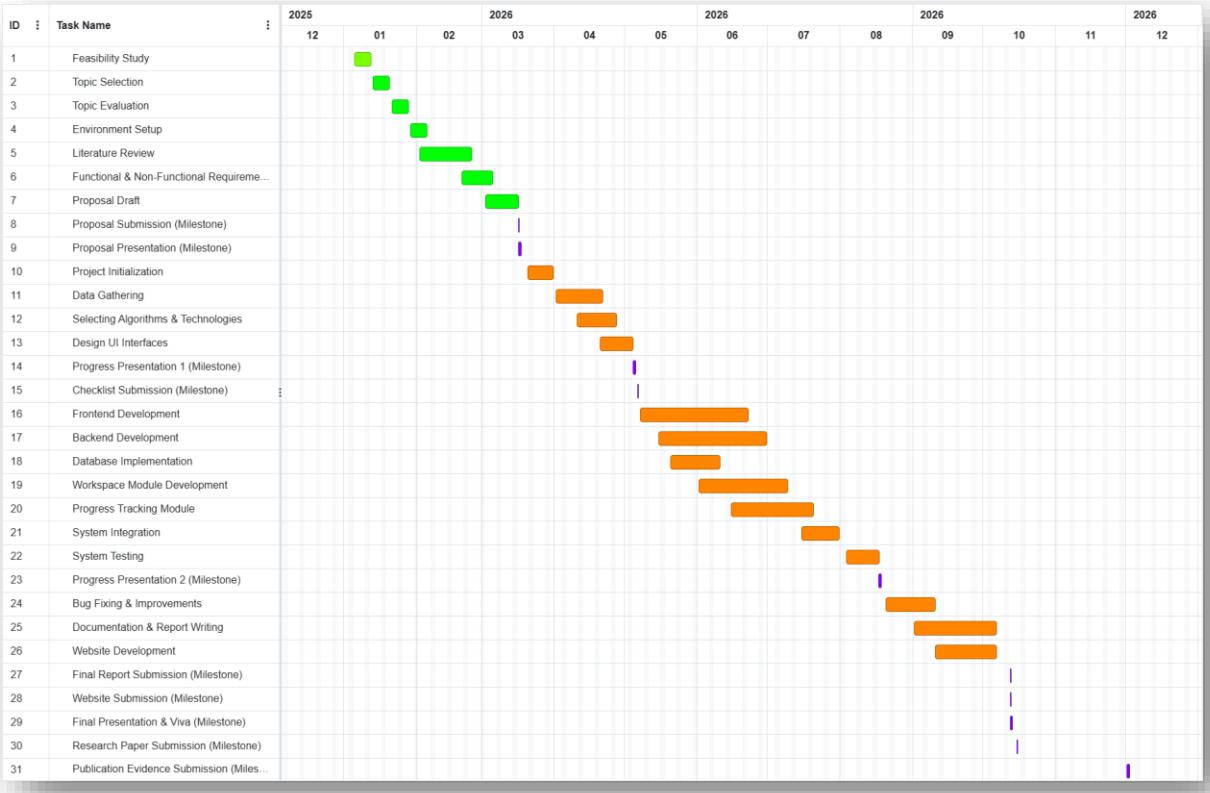


Figure 4 - Grant Chart

10. REFERENCES LIST

[1] V. Arumugam, P. Singh, K. Padhiyar, R. Manek and S. Sayyad, "Academic Project Information Management System," 2021 Asian Conference on Innovation in Technology (ASIANCON), PUNE, India, 2021, pp. 1-6, doi: 10.1109/ASIANCON51346.2021.9544565. keywords: {Technological innovation;Systematics;Automation;Project management;Manuals;User interfaces;Information management;Final year project;Project assessment;Project monitoring;Project management},

[2] J. P. Calvez et al., "Enhancing University Research Portals Through the Application of Knowledge Management Principles," 2025 11th International Conference on Education and Training Technologies (ICETT), Macao, China, 2025, pp. 29-33, doi: 10.1109/ICETT66247.2025.11136912. keywords: {Training;Reviews;Collaboration;Knowledge management;Real-time systems;Research initiatives;Stakeholders;Software tools;Portals;Monitoring;open-access;research management system;university-based research portal;knowledge management principles},

[3] E. N. Desyatirikova, O. V. Minakova, A. S. Golikov, L. V. Chernenkaya and Y. V. Khripunov, "Monitoring Software for Project Based Learning Tasks," 2021 IV International Conference on Control in Technical Systems (CTS), Saint Petersburg, Russian Federation, 2021, pp. 154-157, doi: 10.1109/CTS53513.2021.9562785. keywords: {Databases;Project management;Process control;Memory;Tools;Servers;Software tools;software development;automatic assessment;tracking changes;design stage;status project},

[4] *I. M. Darussalam and Y. Widyani, "GitMonitor: a Software for Monitoring Project Schedules with Git Data," 2021 International Conference on Data and Software Engineering (ICoDSE), Bandung, Indonesia, 2021, pp. 1-5, doi: 10.1109/ICoDSE53690.2021.9648433. keywords: {Schedules;Project management;Feature extraction;Software;Data mining;Synchronization;Stakeholders;schedule;Git;GitMonitor;task;branch},*

[5] *B. Scott-Hill, C. Anslow, J. Ferreira, M. Kropp, M. Mateescu and A. Meier, "Visualizing Progress Tracking for Software Teams on Large Collaborative Touch Displays," 2020 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC), Dunedin, New Zealand, 2020, pp. 1-5, doi: 10.1109/VL/HCC50065.2020.9127286. keywords:*

{Visualization;Collaborative software;Refining;Collaboration;Data visualization;Agile software development;Touch sensitive screens;Collaborative Software Development;Visualization;Software Teams;Progress Tracking},

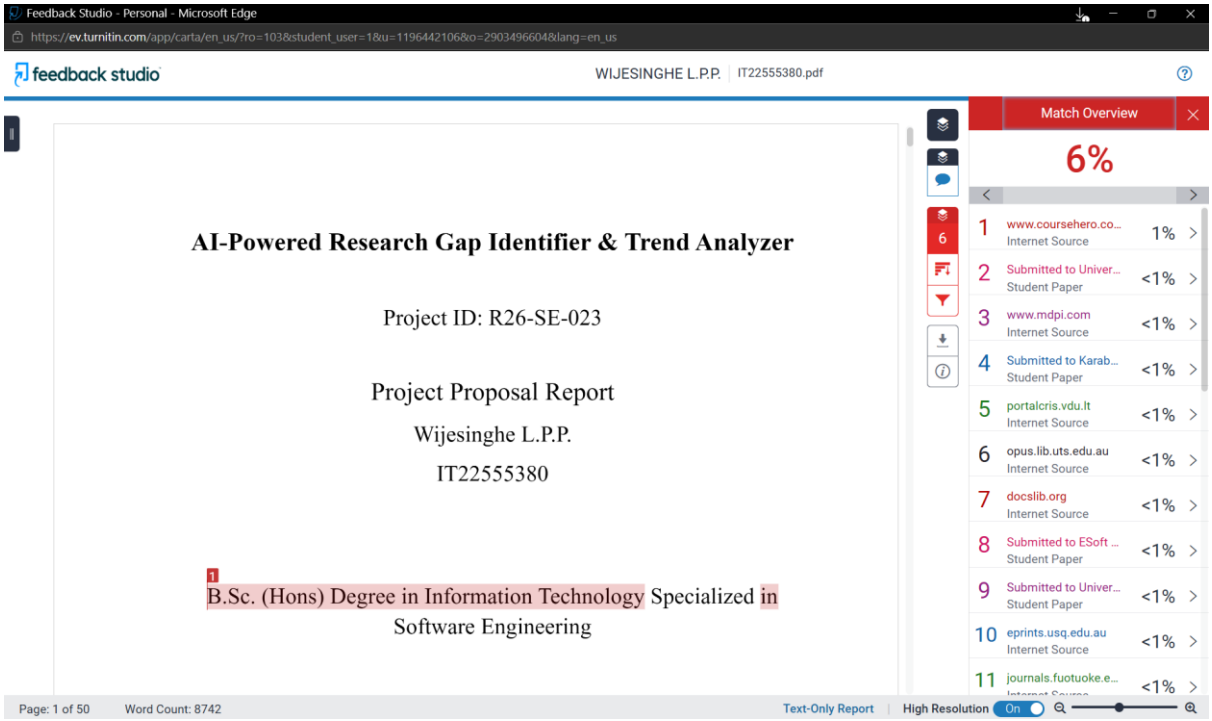
[6] E. Tüzün, Ç. Üsfekes, Y. Macit and G. Giray, "Towards Unified Software Project Monitoring for Organizations using Hybrid Processes and Tools," 2019 IEEE/ACM International Conference on Software and System Processes (ICSSP), Montreal, QC, Canada, 2019, pp. 115-119, doi: 10.1109/ICSSP.2019.00023. keywords: {Organizations;Manuals;Software;Kernel;Monitoring;Software development management;hybrid processes, hybrid tools, software process, project monitoring, project tracking, essence framework},

[7] *L. Horváth, "Model Organized Theoretical and Experimental Research in Collaborative Space," 2023 IEEE 17th International Symposium on Applied Computational Intelligence and Informatics (SACI), Timisoara, Romania, 2023, pp. 000247-000252, doi: 10.1109/SACI58269.2023.10158625. keywords: {Knowledge engineering;Cloud computing;Renewable energy sources;Computational modeling;Collaboration;Organizations;Market research;research in collaborative space;experimental model for research;research project;engineering modeling platform},*

[8] *H. Alaidaros, M. Omar, R. Romli and A. Hussein, "The Development and Evaluation of A Progress Monitoring Prototype Tool for Software Project Management," 2019 First International Conference of Intelligent Computing and Engineering (ICOICE), Hadhramout, Yemen, 2019, pp. 1-9, doi: 10.1109/ICOICE48418.2019.9035146. keywords: {Tools;Monitoring;Software;Task analysis;Prototypes;Project management;Visualization;software project management;Agile methods;Kanban method;progress monitoring;PM-PT},*

[9] *R. Mirza, K. M. Anderson and S. Voida, "EPIC Collab: Supporting Asynchronous Collaboration in Big Data Analysis Systems," 2021 IEEE 6th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA), Chengdu, China, 2021, pp. 17-26, doi: 10.1109/ICCCBDA51879.2021.9442495. keywords: {Cloud computing;Data analysis;Social networking (online);Conferences;Collaboration;Big Data;Software systems;groupware;collaborative analysis;asynchronous collaboration;big data;crisis informatics;software system},*

11. APPENDICES



APPENDICES 1- Plagiarism Report